

Dietary supplementation with flaxseed oil lowers blood pressure in dyslipidaemic patients.

OBJECTIVE: Alpha-linolenic acid (ALA) is the natural precursor of the cardioprotective long-chain n-3 fatty acids. Available data indicate a possible beneficial effect of ALA on cardiovascular disease (CVD), but the response of various CVD risk factors to increased ALA intake is not well characterized. The purpose of the present study was to examine the effect of increased ALA intake on blood pressure in man.

DESIGN, SETTING, SUBJECTS AND INTERVENTIONS: We used a prospective, two-group, parallel-arm design to examine the effect of a 12-week dietary supplementation with flaxseed oil, rich in ALA (8 g/day), on blood pressure in middle-aged dyslipidaemic men (n=59). The diet of the control group was supplemented with safflower oil, containing the equivalent n-6 fatty acid (11 g/day linoleic acid (LA); n=28). Arterial blood pressure was measured at the beginning and at the end of the dietary intervention period.

RESULTS: Supplementation with ALA resulted in significantly lower systolic and diastolic blood pressure levels compared with LA ($P=0.016$ and $P=0.011$, respectively, from analysis of variance (ANOVA) for repeated measures).

CONCLUSIONS: We observed a hypotensive effect of ALA, which may constitute another mechanism accounting in part for the apparent cardioprotective effect of this n-3 fatty acid.